

Aptitude formulas and shortcuts

For students preparing for campus online tests (TCS NQT, Infosys, AMCAT-style). Quantitative formulas with a worked example for each topic, logical reasoning and data interpretation tips, and a time plan for test day.

How to use this guide

- **Learn the formula, then do the example yourself** before you read the answer. Then cover the answer and do it again the next day.
- **Speed matters more than difficulty.** Most campus aptitude questions are easy to medium, but you get only about 1 to 1.5 minutes each. Shortcuts save that time.
- **Write units and what "x" means** on your rough sheet. Most wrong answers come from mixing up km/h and m/s, or CP and SP, not from hard maths.
- Notation: **r%** means rate in percent, **CP** = cost price, **SP** = selling price, **MP** = marked price.

Numbers worth memorising

Fraction	%	Fraction	%	Fraction	%
1/2	50	1/7	14.29	1/12	8.33
1/3	33.33	1/8	12.5	1/15	6.67
1/4	25	1/9	11.11	1/16	6.25
1/5	20	1/10	10	1/20	5
1/6	16.67	1/11	9.09	1/25	4

Squares 11 to 25: 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625.

Cubes 1 to 12: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728.

1. Percentages

Situation	Formula
x% of y	$x \times y / 100$
Percentage change	$(\text{new} - \text{old}) / \text{old} \times 100$
Two successive changes a% and b%	Net % = $a + b + ab/100$ (use a minus sign for a decrease)
A is r% more than B; how much less is B than A?	$r / (100 + r) \times 100 \%$
A is r% less than B; how much more is B than A?	$r / (100 - r) \times 100 \%$
Price rises r%; cut consumption by how much to keep spending the same?	$r / (100 + r) \times 100 \%$
Population growing r% a year for n years	$P \times (1 + r/100)^n$

Example: A price rises by 20% and then falls by 10%. What is the net change?

Net = $20 + (-10) + (20 \times -10)/100 = 20 - 10 - 2 = \mathbf{8\% \text{ increase}}$. Check: $100 \rightarrow 120 \rightarrow 108$.

Tip: Take 100 as the starting value whenever the question gives no actual number. It turns every percentage into a simple amount.

2. Profit, loss and discount

Quantity	Formula
Profit / Loss	Profit = $SP - CP$; Loss = $CP - SP$
Profit % / Loss %	Always on CP: Profit / $CP \times 100$
SP from CP	$SP = CP \times (100 + \text{gain}\%) / 100$, or $CP \times (100 - \text{loss}\%) / 100$
CP from SP	$CP = SP \times 100 / (100 + \text{gain}\%)$
Discount %	Always on MP: Discount / $MP \times 100$; $SP = MP \times (100 - d\%) / 100$
Successive discounts a% and b%	Single discount = $a + b - ab/100$
Dishonest dealer (sells at CP, false weight)	Gain % = $\text{error} / (\text{true value} - \text{error}) \times 100$
Two items sold at the same SP, one at x% gain, one at x% loss	Always a loss of $x^2/100 \%$

Example: A shopkeeper marks goods 40% above cost and gives a 10% discount. Find the profit %.

Take $CP = 100$. $MP = 140$. $SP = 140 \times 0.9 = 126$. Profit = **26%**.

Tip: Profit % uses CP, discount % uses MP. Mixing the two bases is the most common mistake in this topic.

3. Simple and compound interest

Quantity	Formula
Simple interest	$SI = P \times R \times T / 100$; Amount = $P + SI$
Compound amount (yearly)	$A = P(1 + R/100)^n$; $CI = A - P$
Half-yearly / quarterly	Use $R/2$ and $2n$, or $R/4$ and $4n$
CI – SI for 2 years	$P(R/100)^2$
CI – SI for 3 years	$P(R/100)^2 \times (3 + R/100)$
Sum doubles under SI	$T = 100 / R$ years
Rule of 72 (CI, approximate)	Doubling time $\approx 72 / R$ years

Example: Find the difference between CI and SI on Rs 10,000 at 10% per year for 2 years.

Shortcut: $10,000 \times (10/100)^2 = \text{Rs } 100$. Check: $CI = 12,100 - 10,000 = 2,100$; $SI = 2,000$; difference = 100.

4. Ratio, proportion and partnership

- If $a : b = c : d$, then **$ad = bc$** .
- Fourth proportional to $a, b, c = \text{bc} / \text{a}$. Third proportional to $a, b = \text{b}^2 / \text{a}$. Mean proportional of $a, b = \sqrt{ab}$.
- Divide an amount S in ratio $a : b \rightarrow S \times a/(a + b)$ and $S \times b/(a + b)$.
- Combine $a : b$ and $b : c$ by making the common term equal (multiply so that both "B" values match).
- **Partnership:** profit is shared in the ratio of (capital $\times$ time).

Example: $A : B = 2 : 3$ and $B : C = 4 : 5$. Divide Rs 7,000 among A, B and C.

Make B equal: $2 : 3 = 8 : 12$ and $4 : 5 = 12 : 15$. So $A : B : C = 8 : 12 : 15$ (total 35). One part = $7,000 / 35 = 200$. Shares = **Rs 1,600, Rs 2,400, Rs 3,000**.

5. Averages

- Average = sum of values / number of values. So **sum = average $\times$ count**; most questions are solved with sums, not averages.
- Average of first n natural numbers = $(n + 1)/2$. For consecutive numbers (or any equally spaced list), average = middle term = $(\text{first} + \text{last})/2$.

- **Replacement:** if the average of n people rises by d when a person of value w is replaced, the new person = $w + n \times d$ (falls by d : $w - n \times d$).
- **Weighted average:** $(n_1a_1 + n_2a_2) / (n_1 + n_2)$.

Example: The average weight of 8 people rises by 2.5 kg when a person weighing 65 kg is replaced by a new person. Find the new person's weight.

Total rise = $8 \times 2.5 = 20$ kg. New person = $65 + 20 = 85$ kg.

6. Mixtures and alligation

- **Alligation rule:** Quantity of cheaper : Quantity of dearer = (dearer price – mean price) : (mean price – cheaper price). Works for prices, percentages, concentrations and speeds.
- **Repeated replacement:** a container holds C litres of pure liquid. x litres are taken out and replaced with water, n times. Pure liquid left = $C \times (1 - x/C)^n$.

Example: In what ratio must rice at Rs 40/kg be mixed with rice at Rs 55/kg to get a mixture worth Rs 50/kg?

Cheaper : dearer = $(55 - 50) : (50 - 40) = 5 : 10 = 1 : 2$.

Tip: Draw the alligation cross (two prices on top, mean in the middle, differences below). It takes five seconds and avoids sign errors.

7. Time and work

Idea	Formula
A finishes the work in a days	A's one-day work = $1/a$
A and B together	$1/a + 1/b$, so time = $ab / (a + b)$
LCM method	Total work = LCM of the days; each person's per-day work = total / their days
Men, days, hours	$M_1D_1H_1 / W_1 = M_2D_2H_2 / W_2$
Efficiency	If A is k times as efficient as B, A takes $1/k$ of B's time

Example: A can do a job in 12 days and B in 18 days. How long will they take together?

Total work = LCM(12, 18) = 36 units. A does 3 units/day, B does 2 units/day, together 5 units/day. Time = $36 / 5 = 7.2$ days.

Tip: The LCM method keeps everything in whole numbers. Use it for any question with three or more people, or people leaving midway.

8. Pipes and cisterns

- Same as time and work. An inlet pipe adds work (+), an outlet or leak removes it (–).
- Pipe fills in a hours, another empties in b hours ($b > a$): both open, tank fills in $\frac{ab}{b - a}$ hours.
- A pipe fills in a hours, but with a leak it takes b hours: the leak alone empties a full tank in $\frac{ab}{b - a}$ hours.

Example: Pipes A and B fill a tank in 10 h and 15 h. Pipe C empties it in 30 h. If all three are opened together, when is the tank full?

Take tank = 30 units (LCM). A = +3, B = +2, C = –1 per hour. Net = 4 units/hour. Time = $30 / 4 = 7.5$ hours.

9. Time, speed and distance

Idea	Formula
Basic	Distance = Speed $\times$ Time
Unit change	km/h $\rightarrow$ m/s: $\times 5/18$. m/s $\rightarrow$ km/h: $\times 18/5$
Average speed over two equal distances at x and y	$2xy / (x + y)$ (not $(x + y)/2$)
Same distance, speed ratio a : b	Time ratio b : a
Relative speed	Opposite directions: add speeds. Same direction: subtract.

Trains

- Train of length L passes a pole or a person: time = L / speed .
- Passes a platform or bridge of length P: time = $(L + P) / \text{speed}$.
- Two trains, lengths L_1, L_2 : time to cross = $(L_1 + L_2) / (s_1 + s_2)$ if moving towards each other, or $(s_1 - s_2)$ if moving the same way.

Example: A 150 m train running at 72 km/h crosses a 250 m platform. How long does it take?

Speed = $72 \times 5/18 = 20$ m/s. Distance = $150 + 250 = 400$ m. Time = $400 / 20 = 20$ seconds.

Boats and streams

- Boat speed in still water u, stream speed v. Downstream = $u + v$. Upstream = $u - v$.
- $u = (\text{downstream} + \text{upstream}) / 2$. $v = (\text{downstream} - \text{upstream}) / 2$.

Example: A man rows at 15 km/h in still water. The stream flows at 3 km/h. How long does he take to go 36 km downstream and come back?

Downstream speed 18 km/h: $36/18 = 2$ h. Upstream speed 12 km/h: $36/12 = 3$ h. Total = **5 hours**.

Tip: Convert all units first. If the answer options are in seconds and the speed is in km/h, change the speed to m/s before anything else.

10. Number system

Divisibility rules

By	Rule	By	Rule
2	Last digit even	8	Last three digits divisible by 8
3	Digit sum divisible by 3	9	Digit sum divisible by 9
4	Last two digits divisible by 4	10	Last digit 0
5	Last digit 0 or 5	11	(Sum of odd-place digits) – (sum of even-place digits) is 0 or a multiple of 11
6	Divisible by both 2 and 3	12	Divisible by both 3 and 4
7	Double the last digit and subtract it from the rest of the number; repeat. The number is divisible by 7 if the result is (e.g. 343: $34 - 6 = 28$, divisible by 7).		

LCM, HCF and other facts

- For two numbers: **LCM $\times$ HCF = product of the numbers.**
- HCF of fractions = HCF of numerators / LCM of denominators. LCM of fractions = LCM of numerators / HCF of denominators.
- Smallest number that leaves the **same remainder r** when divided by $a, b, c = \text{LCM}(a, b, c) + r$.
- If each (divisor – remainder) is the same value k , the smallest such number = LCM – k . Example: divided by 4, 5, 6 leaves 3, 4, 5. Here $k = 1$, so the answer is $60 - 1 = 59$.
- Number of factors:** if $N = p^a \times q^b \times r^c$ (primes), factors = $(a+1)(b+1)(c+1)$. Example: $72 = 2^3 \times 3^2$, so $4 \times 3 = 12$ factors.
- Trailing zeros in $n!$** = $n/5 + n/25 + n/125 + \dots$ (take only the whole-number part of each). $100!$ has $20 + 4 = 24$ zeros.
- Sums: $1 + 2 + \dots + n = n(n+1)/2$. Squares: $n(n+1)(2n+1)/6$. Cubes: $[n(n+1)/2]^2$.

Remainders and unit digits

- Remainder of a product (or sum) = product (or sum) of the individual remainders, then take the remainder again.
- Find a power that leaves remainder 1 (or -1), then break the big power around it.
- Unit-digit cycles:** $2 \rightarrow 2, 4, 8, 6$; $3 \rightarrow 3, 9, 7, 1$; $7 \rightarrow 7, 9, 3, 1$; $8 \rightarrow 8, 4, 2, 6$ (cycle of 4). $4 \rightarrow 4, 6$ and $9 \rightarrow 9, 1$ (cycle of 2). 0, 1, 5, 6 never change. Divide the power by 4: remainder 1, 2, 3 gives the 1st, 2nd, 3rd term; remainder 0 gives the 4th.

Example: Find the remainder when 2^{100} is divided by 7.

$2^3 = 8$ leaves remainder 1 on division by 7. $100 = 3 \times 33 + 1$, so $2^{100} = (2^3)^{33} \times 2$ leaves $1^{33} \times 2 = 2$.

11. Permutations and combinations

Idea	Formula
Arrange r out of n (order matters)	${}^n P_r = n! / (n - r)!$
Choose r out of n (order does not matter)	${}^n C_r = n! / (r! (n - r)!)$; also ${}^n C_r = {}^n C_{n-r}$
Arrange n items with p alike, q alike	$n! / (p! q!)$
n people around a round table	$(n - 1)!$; necklace or garland (flip allowed): $(n - 1)! / 2$
Choose at least one from n items	$2^n - 1$
Handshakes among n people / lines from n points (no three in a line)	${}^n C_2 = n(n - 1)/2$
Diagonals of an n-sided polygon	$n(n - 3)/2$

- **AND** → **multiply**, **OR** → **add**. "Choose 2 men and 1 woman" multiplies; "2 men or 2 women" adds.
- "Items always together": tie them as one unit, then multiply by their internal arrangements.
- "At least" questions: often total minus the unwanted cases is faster.

Example: A committee of 3 men and 2 women is to be formed from 6 men and 5 women. In how many ways?

$${}^6 C_3 \times {}^5 C_2 = 20 \times 10 = \mathbf{200 \text{ ways.}}$$

12. Probability

- $P(E)$ = favourable outcomes / total outcomes, between 0 and 1.
- $P(\text{not } E) = 1 - P(E)$. Best for "at least one" questions.
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Independent events: $P(A \text{ and } B) = P(A) \times P(B)$.
- **Coins:** n tosses give 2^n outcomes. **Dice:** n dice give 6^n outcomes.
- **Cards:** 52 cards, 4 suits of 13 (spades and clubs black; hearts and diamonds red), 26 red, 26 black, 12 face cards (J, Q, K), 4 aces.

Sum of two dice	2	3	4	5	6	7	8	9	10	11	12
Ways (out of 36)	1	2	3	4	5	6	5	4	3	2	1

Example: Two dice are thrown. What is the probability that the sum is 8?

Favourable: (2,6), (3,5), (4,4), (5,3), (6,2) = 5. Total = 36. $P = \mathbf{5/36}$.

Tip: "At least one head in 3 tosses" = $1 - P(\text{no head}) = 1 - 1/8 = 7/8$. Use the complement whenever you see "at least".

13. Problems on ages

- Let the present age be x . "n years ago" = $x - n$; "after n years" = $x + n$.
- The **difference** between two people's ages never changes.
- For ratio questions, write present ages as ak and bk , form one equation, solve for k .
- Quick check: put each option into the question. With 4 options this is often faster than algebra.

Example: The ages of A and B are in the ratio 4 : 5. After 6 years the ratio will be 5 : 6. Find their present ages.

$(4k + 6) / (5k + 6) = 5/6 \rightarrow 24k + 36 = 25k + 30 \rightarrow k = 6$. A = **24**, B = **30**. Check: after 6 years, $30 : 36 = 5 : 6$.

14. Clocks and calendars

Clocks

- Minute hand moves 6° per minute; hour hand moves 0.5° per minute. Minute hand gains 5.5° per minute (55 minute-spaces per hour).
- **Angle between hands at H:M** = $|30H - 5.5M|$. If the result is more than 180° , subtract it from 360° .
- Hands coincide between H and H+1 o'clock at **60H/11 minutes** past H (e.g. between 3 and 4: $180/11 = 16 \frac{4}{11}$ min past 3).
- In 12 hours, the hands coincide 11 times, are opposite (180°) 11 times and are at right angles 22 times. Double these for a full day (22, 22, 44).

Example: What is the angle between the hands at 3:40?

$|30 \times 3 - 5.5 \times 40| = |90 - 220| = \mathbf{130^\circ}$.

Calendars

- **Odd days** = days left over after complete weeks. Ordinary year = 1 odd day, leap year = 2.
- **Leap year:** divisible by 4; a century year (1900, 2000) must be divisible by 400. So 2000 was a leap year, 1900 was not.
- Odd days in centuries: 100 years = 5, 200 years = 3, 300 years = 1, 400 years = 0.
- Months: 31 days = 3 odd days, 30 days = 2, February = 0 (or 1 in a leap year).
- Day codes: 0 = Sunday, 1 = Monday, ... 6 = Saturday.
- Same calendar repeats (no skipped century leap year in between): leap year $\rightarrow$ after 28 years; year after a leap year $\rightarrow$ after 6 years; second or third year after a leap year $\rightarrow$ after 11 years.

Example: What day of the week was 15 August 1947?

Complete years up to 1946: 1600 years = 0 odd days; next 300 years = 1; 46 years = 11 leap + 35 ordinary = $22 + 35 = 57 = 1$ odd day. Total so far = 2.

1947, 1 Jan to 15 Aug: Jan 3, Feb 0, Mar 3, Apr 2, May 3, Jun 2, Jul 3, Aug 15 days = 1. Sum = $17 = 3$ odd days.

Total = $2 + 3 = 5 \rightarrow$ **Friday**.

Logical reasoning tips

Number and letter series

- Check in this order: differences, second differences, ratios, squares/cubes $\pm$ something, primes, two series mixed together (odd and even positions).
- Example: 2, 6, 12, 20, 30, ? $\rightarrow$ differences 4, 6, 8, 10, next 12 $\rightarrow$ **42** (also $n(n+1)$).
- Letter positions: A = 1 ... Z = 26. Remember **EJOTY** = 5, 10, 15, 20, 25. Reverse position of a letter = $27 -$ its position ($A \leftrightarrow Z$, $B \leftrightarrow Y$).

Coding-decoding

- Write the letter positions under both the word and its code; the pattern (+1, -2, reverse, alternate) usually shows at once.
- Example: if CAT is written DBU (each letter +1), then DOG is written **EPH**.
- For "word codes" (sentences coded as words), find the word common to two sentences; its code is the code common to both.

Blood relations

- Always draw a family tree: + for male, - for female, horizontal line for couple, vertical line for parent-child.
- Do not assume gender from a name. "A is the child of B" does not tell you A's gender.
- Start decoding from the far end of a sentence like "A is the son of the brother of B's father".

Seating arrangement

- Place the person with the most definite clue first, then fill neighbours. Draw several possible cases if needed and remove the ones a later clue breaks.
- Linear row facing north: their left is your left. Facing south: reversed.
- Circle facing the centre: left = clockwise, right = anticlockwise. Facing outward: reversed.

Syllogisms

- Draw the smallest Venn diagram that fits each statement. A conclusion is true only if it holds in **every** possible diagram.
- "All A are B" does **not** mean "All B are A", but it does give "Some B are A".
- "Some A are B" = "Some B are A". "No A is B" = "No B is A".
- Two "some" statements give no definite conclusion between the end terms.
- Example: All cats are dogs. All dogs are birds. Then "All cats are birds" and "Some birds are cats" both follow. "All birds are cats" does not.
- "Either-or" option: pick it when two conclusions about the same pair cannot both be true but one must be (e.g. "Some A are B" and "No A is B", when neither follows alone).

Direction sense

- Draw N at the top every time. A right turn when facing north means facing east. Final distance from start uses Pythagoras: $\sqrt{x^2 + y^2}$.

Data interpretation tips

- **Read the question first**, then the chart. Note units (thousands, lakhs, crores, %) and what the totals mean.
- **Approximate**. If options are far apart, round numbers (e.g. $4,987 \approx 5,000$). Only calculate exactly when options are close.
- Use fraction-percent values: $37.5\% = 3/8$, $62.5\% = 5/8$, $87.5\% = 7/8$, $16.67\% = 1/6$.
- **Pie charts**: value = (sector angle / 360) $\times$ total, or (sector % / 100) $\times$ total.
- **Growth**: % change = (this year – last year) / last year $\times$ 100. The base is always the earlier value.
- For "which year had the highest growth" questions, compare ratios (new/old) rather than computing every percentage.
- Ratio questions often do not need the total. Cancel common factors before you multiply.
- One DI set usually has 4 to 5 questions on the same data, so spending a minute to understand the chart pays back.

Tip: Write down intermediate totals you compute (a column sum, a yearly total). The next question in the set often needs the same number.

Time-management strategy for the test

1. **Read the instructions screen carefully.** Check whether sections have separate timers, whether there is negative marking, whether you can go back to a question, and whether an on-screen calculator is allowed. These differ between tests and between years.
2. **Two-pass method.** First pass: answer every question you can solve in under a minute. Skip or mark the rest. Second pass: return to marked questions with the time left.
3. **The 90-second rule.** If you have no clear method after about 90 seconds, move on. One stuck question can cost you three easy ones.
4. **Use the options.** Eliminate options with the wrong unit digit, wrong parity, or a value that is obviously too large or small. Put options back into the question for ages, equations and number questions.
5. **No negative marking?** Then never leave a question blank. Guess after eliminating what you can. With negative marking, guess only if you can remove at least two options.
6. **Keep your rough sheet tidy.** Number every rough calculation with its question number so you can recheck quickly.
7. **Watch the clock at fixed points** (for example, at the halfway mark of each section) instead of after every question.

Suggested order inside a quant section

Do first (fast, high accuracy)	Do next	Leave for the end
Percentages, profit and loss, SI/CI, averages, ratio, ages, simple number system	Time and work, pipes, TSD, trains, boats, mixtures, clocks, calendars	Long P&C, tricky probability, multi-step remainders, lengthy DI sets

Last-week checklist

- Revise this formula sheet once a day; say each formula aloud without looking.
 - Take at least 3 full-length timed mock tests in the same pattern as your target test, and review every wrong answer.
 - Keep an error log: topic, what went wrong (concept, calculation, misreading), and the correct method.
 - Practise mental maths daily for 10 minutes: squares, percentages of numbers, quick division.
 - Revisit your two weakest topics only; do not start completely new topics in the last two days.
 - Check test-day details: exam slot, ID proof, admit card, and a stable internet connection and working webcam if the test is remote.
 - Sleep well the night before. A fresh mind is faster than one extra hour of revision.
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